DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics III Topics in Analysis (3TN)

Tutorial Sheet 6 **16 September 2024**

1. Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be two normed spaces over the field $\mathbb{R}$ and Ω be a non-empty open subset of V.

   (a) Define what is meant by the statement that the mapping $f : \Omega \to W$ is Gâteaux differentiable at $x_0 \in \Omega$ and define the Gâteaux derivative G_{x_0} of the function f at x_0.

   (b) Define what is meant by the statement that the mapping $f : \Omega \to W$ is Fréchet differentiable at $x_0 \in \Omega$ and define the Fréchet derivative $Df(x_0)$ of the function f at x_0.

   (c) Show that if $f : \Omega \to W$ is Fréchet differentiable at x_0 with Frévhet derivative $Df(x_0)$, then f is also Gâteaux differentiable at x_0 with Gâteaux derivative $G_{x_0} = Df(x_0)$.

   (d) Show that if f is Fréchet differentiable at x_0, then its Fréchet derivative $Df(x_0)$ at x_0 is unique.

2. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x,y) := \begin{cases} \dfrac{y^2 \sin^2(x)}{x^2+y^2} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0)\,. \end{cases}$$

   (i) Find the directional derivative $f_v(0,0)$ of the function f at $(0,0)$ in the direction of the vector $v = (v_1, v_2) \neq (0,0)$.

   (ii) Say whether the function f is Gâteaux differentiable at $(0,0)$.

   (iii) Is the function f Fréchet differentiable at the point $(0,0)$? Explain your answer!

3. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x,y) := \begin{cases} \dfrac{y^3}{x^2+y^2} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0)\,. \end{cases}$$

   (i) Find the directional derivative $f_v(0,0)$ of the function f at $(0,0)$ in the direction of the vector $v = (v_1, v_2) \neq (0,0)$.

   (ii) Deduce that the function f is not Fréchet differentiable at the point $(0,0)$.

4. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x,y) := \begin{cases} \dfrac{\sin\left(\sqrt{x^2+y^2}\right)}{\sqrt{x^2+y^2}} & \text{if } (x,y) \neq (0,0), \\ \\ 1 & \text{if } (x,y) = (0,0)\,. \end{cases}$$

   (a) Show that f is continuous at $(0,0)$.

   (b) Establish whether the function f is Fréchet differentiable at $(0,0)$ and find (if possible) $Df(0,0)$ (the Fréchet derivative at $(0,0)$).

5. Let $\mathbb{R}_{\leq n}[X]$ be the vector space of polynomials of degree less or equal to n and let

$$V := \left\{ P|_{[0,1]} \colon\, P \in \mathbb{R}_{\leq n}[X] \right\}$$

   be the vector space of restrictions to $[0,1]$ of polynomials of degree less or equal to n, equipped with norm $\|\cdot\|_\infty$ defined by

$$\|P\|_\infty := \sup_{t\in[0,1]} |P(t)| \quad \forall P \in V\,.$$

   Study the Fréchet differentiability of the mapping $f_1,\ f_2 : V \to \mathbb{R}$ defined by

$$f_1(P) = \int_0^1 [(P(x))^2]dx \quad \text{and} \quad f_2(P) = \int_0^1 [(P(x))^3]dx \qquad \forall P \in V$$

   and find their Fréchet derivatives $Df_1(P)$ and $Df_2(P)$ at every $P \in V$.

6. Give an example of function $f : \mathbb{R}^2 \to \mathbb{R}$ continuous and Gâteaux differentiable at $(0,0)$, but not Fréchet differentiable at that point.